Julia + Jupyter Environment Setup (Issue #1)

October 14, 2026

Decision

Docker, inside the existing project container (`opt`), is the primary interface: it follows the repo rule of a single container, serves JupyterLab at `localhost:8896`, and is reproducible. The base image only ships Julia 1.9, so `code/Dockerfile` now installs Julia 1.12.5 (official tarball, architecture-aware) with its depot in `/opt/julia_depot`, plus IJulia. Packages live in a project-local environment, `code/scripts/julia/` (`Project.toml`, `Manifest.toml`), selected by the kernel `julia-project` ("Julia 1.12 (project)").

Verification

The image builds; `jupyter kernelspec list` shows `julia-project`; a notebook executed headlessly with that kernel reports Julia 1.12.5 with active project `/home/repo-intro/scripts/julia/Project.toml`; `Pkg.add("DataFrames")` installed and loaded (test change reverted, so the committed environment is empty).

Native alternative

juliaup (Julia 1.12.5 already installed on the host) with VS Code's Julia extension is the best native interface; the same `Manifest.toml` is used. Details in `code/scripts/julia/README.md` and the repository `README.md`.

Follow-ups (not done)

Starter notebook; pin the Docker base image (`jupyter/datascience-notebook:latest` on Docker Hub is stale).